

## Supplementary Requirements Documentation

**Title of Project: Interactive House - AI smart control system**

**Project's Github repo link:**

<https://github.com/DanielJ0131/interactive-house>

### Revision History

| Date       | Version | Description                                                                                                                         | Author                        |
|------------|---------|-------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| 2026-02-10 | 1.0     | Initial supplementary requirements for Interactive House                                                                            | Anitta Antony                 |
| 2026-02-24 | 1.1     | Refined supplementary requirements aligned with AI-based system (S1-S4)                                                             | Md Abu Taher                  |
| 2026-02-24 | 1.1     | Refined supplementary requirements aligned with AI-based system (S5-S7)                                                             | Anitta Antony                 |
| 2026-02-28 | 1.0     | Smart Home code                                                                                                                     | Daniel Marcarini              |
| 2026-02-28 | 1.0     |                                                                                                                                     | Daniel Holderik               |
| 2026-03-16 | 1.2     | Update current requirements with Firebase AI and web                                                                                | Anitta Antony<br>Md Abu Taher |
| 2026-03-16 | 1.1     | Testing                                                                                                                             | Daniel Marcarini              |
| 2026-04-22 | 1.3     | Added measurable AI usability and performance thresholds, including 1-second feedback and 5-second AI response target               | Anitta Antony<br>Md Abu Taher |
| 2026-05-08 | 1.4     | Updated final progress status and clarified supplementary requirements for AI, security, documentation quality, and web testability | Anitta Antony<br>Md Abu Taher |
| 2026-05-08 | 1.5     | Updated S8 with current Jest automated testing and Istanbul coverage status                                                         | Daniel Marcarini              |
| 2026-05-21 | 1.6     | Updated S1 and S8 with browser speech recognition                                                                                   | Daniel Holderik               |
| 2026-05-23 | Final   | Updated final testing status and project completion evidence                                                                        | Daniel Marcarini              |

### Supplementary Requirements List

| Supplementary Requirement Name                                   | Priority  | Progress |
|------------------------------------------------------------------|-----------|----------|
| S1. Usability – Intuitive AI interaction                         | Essential | Done     |
| S2. Reliability – Safety-critical stability                      | Essential | Done     |
| S3. Performance – Real-time responsiveness                       | Essential | Done     |
| S4. Security - Protection of smart devices                       | Essential | Done     |
| S5. Maintainability and scalability                              | Desirable | Done     |
| S6. Design documentation quality                                 | Essential | Done     |
| S7. Technology flexibility                                       | Optional  | Done     |
| S8. Testability - Verifiable web interaction and synchronization | Essential | Done     |

## Supplementary Requirements Descriptions

### S1 Usability – Intuitive AI interaction

To ensure intuitive interaction, the system shall respond to user commands within an acceptable time so that the user does not feel the system is unresponsive. In user interface design, response time is an important factor for usability. According to usability guidelines described by Nielsen (1993), a system should provide feedback within about 1 second to keep the user focused, while delays longer than several seconds may reduce usability.

Based on these guidelines, the system shall display immediate feedback within 1 second after the user sends a command, for example by showing a loading or processing indicator. The final AI response should normally be returned within 5 seconds. Response times longer than this may cause the interaction to feel slow, therefore the system should be designed so that most responses remain within this limit.

These limits are used as the target performance for the AI interaction to ensure that the system remains easy to use and understandable for the user.

The emergency button shall also be visible and easy to access so that users can quickly activate the emergency assistance flow without needing to use AI or speech commands.

The system also supports voice interaction through the Speech APIs. Voice interaction should remain responsive and allow navigation and device-control actions without requiring manual interaction with the interface.

## **S2 Reliability – Safety-critical stability**

Safety-related components shall operate independently of the AI chat interface to ensure system reliability. In safety-critical systems, functions such as alarm activation, sensor monitoring, and automatic responses must not depend on user interaction or external services. According to software engineering guidelines, safety-critical functionality should be separated from non-critical modules so that failures in one part of the system do not affect the operation of essential components (Sommerville, 2016).

Based on this principle, the system shall ensure that sensor monitoring and automatic safety actions are handled directly by the device or controller subsystem without requiring the AI module. The AI interface shall only provide user interaction and must not block or delay safety responses. Safety actions, such as alarm activation or automatic device control, shall occur immediately when triggered by sensor data, even if the AI interface is unavailable.

The emergency assistance feature shall be accessible independently from the AI response process, because urgent user actions should not depend on external AI service availability or response time.

## **S3 Performance – Real time responsiveness**

The system shall provide responsive feedback to the user to maintain smooth interaction. According to usability guidelines, feedback should be visible within about 1 second to keep the user focused, while longer delays require a processing indicator (Nielsen, 1993). Therefore, the system shall display a status or loading message within 1 second after a command is sent.

Because AI processing requires network communication, the final response may take longer, but it should normally be returned within approximately 5 seconds. Response times longer than this may reduce usability, so the system should be designed to keep most responses within this limit. Performance improvements may be applied at the client or AI service level if needed.

## **S4 Security-Protection of smart devices**

The system shall support secure communication between the application and the AI service to prevent unauthorized control of smart devices. Access to device control functions shall require user authentication, and only authorized users shall be allowed to send commands to the system. Security mechanisms such as authentication and access control shall be used to reduce the risk of unintended or malicious actions.

If the user is not authenticated, access to protected device-control pages should be restricted and the user should be redirected to the login flow.

In software systems that control physical devices, access restrictions are important to ensure safe operation and prevent misuse (Sommerville, 2016).

## **S5 Maintainability and scalability**

The system architecture shall be designed to support future expansion, including the addition of new devices, sensors, or AI functions, without requiring major changes to existing components. The design shall follow a modular structure where different parts of the system, such as the user interface, AI module, and device controller, operate independently. This separation of concerns makes the system easier to maintain and extend as new features are added.

Modular design and separation of components are recommended in software engineering to improve maintainability and allow systems to evolve without large redesign (Sommerville, 2016).

## **S6 Design documentation quality**

The system design shall be documented in a clear and consistent way so that the structure and behavior of the system can be understood by developers and reviewers. The documentation shall include requirement specifications, architectural descriptions, and supporting design artifacts such as diagrams or explanations of system components. Proper documentation is required to make future development, maintenance, and extension of the system easier.

Clear and structured documentation is recommended in software engineering to support system understanding and long-term development (Sommerville, 2016).

## **S7 Technology flexibility**

The system shall be designed so that different technologies or AI providers can be used without changing the overall architecture. The AI module, user interface, and device control components shall remain separated so that one part can be replaced without requiring major changes to the rest of the system. This allows the system to use external services or alternative implementations while keeping the main functionality unchanged.

Software engineering guidelines recommend separating system components to allow flexibility and replacement of technologies without redesigning the whole system (Sommerville, 2016).

The system may utilize browser-based APIs such as the Web Speech API, provided compatibility and performance requirements are maintained.

## **S8 Testability – Verifiable web interaction and synchronization**

The system shall support repeatable verification of the main web interaction flows, including authentication, device toggling, and synchronization of state data between the user interface and Firestore. The design shall allow testers to confirm that user actions produce correct backend updates and that backend changes are reflected correctly in the interface.

The web application now includes an initial automated test setup using Jest and React Testing Library. This improves testability by allowing important frontend behavior to be verified repeatedly through automated test execution.

The current automated tests verify login validation, signup validation, successful authentication-related flows, Firebase Auth configuration, and Firestore configuration. Firebase services are mocked during testing, which allows the tests to run without modifying real backend data.

The current testability status is partially complete. Authentication and Firebase configuration are covered by automated tests, while hub device toggles, Firestore live synchronization, emergency flow, music functionality, and AI interaction still require additional automated test coverage or manual verification.

The web application additionally supports browser-based voice interaction using the Web Speech API. Current voice functionality includes transcript generation, page navigation commands, and device-control commands integrated with the existing Firebase-backed control flow. Voice interaction currently depends on browser compatibility and is primarily supported in Chromium-based browsers.

### **Final Testability Status**

S8 is fulfilled for the final web application scope. The project now includes repeatable automated testing using Jest and React Testing Library.

The final automated test suite includes 49 passing tests across 9 test suites. The tests cover authentication, Firebase configuration, shared UI components, static pages, guest hub, emergency flow, hub device controls, Firestore update behavior, and music page behavior.

The use of mocked Firebase services and mocked browser APIs improves testability because the tests do not depend on real backend state, real user accounts, or real browser audio services.

## **Relation with Requirements, Design, and Test Plan**

The supplementary requirements support the functional requirements by defining quality expectations for usability, reliability, performance, security, maintainability, documentation, technology flexibility, and testability. S1 and S3 are connected to the AI interaction requirements because they define the expected user feedback and AI response-time targets. S2 is connected to the safety-related requirements because emergency and sensor-based responses should not depend on the AI module. S4 supports the device-control requirements by requiring authentication and protected access to smart device functions. S5, S6, and S7 support the design artefact by requiring modularity, clear documentation, and flexibility for future AI or technology changes. S8 is connected to the current web test plan because it supports verification of login, signup, guest access, hub loading, Firestore-backed device toggles, and real-time synchronization.

## Daniel Marcarini - Smart Home code

| Supplementary Requirement Name                          | Priority  |
|---------------------------------------------------------|-----------|
| S1. Usability – Clear physical interaction and feedback | Essential |
| S2. Reliability – Continuous safety monitoring          | Essential |
| S3. Performance – Real-time embedded responsiveness     | Essential |
| S4. Safety - Fail-safe emergency behavior               | Essential |
| S5. Maintainability - Modular embedded design           | Desirable |
| S6. Power stability and hardware robustness             | Essential |
| S7. System documentation quality                        | Essential |

### **S1 Usability – Clear physical interaction and feedback**

The system shall provide intuitive physical interaction through push buttons and clear visual feedback via the LCD display. Users shall be able to control devices without technical knowledge of the internal system architecture. System feedback shall be clear and understandable during both normal and emergency operation.

### **S2 Reliability – Continuous safety monitoring**

The system shall operate reliably under continuous execution. Gas detection and environmental monitoring shall function independently of user interaction. Safety-related monitoring shall remain active at all times during system operation. The system shall not enter undefined states during normal use.

### **S3 Performance – Real-time embedded responsiveness**

The system shall respond to sensor changes and user inputs within an acceptable real-time threshold. Emergency conditions such as gas detection shall trigger immediate response procedures. Timing behavior shall be managed using non-blocking mechanisms to ensure stable concurrent operation of all components.

### **S4 Safety – Fail-safe emergency behavior**

The system shall prioritize safety-related actions over non-critical functionality. When a hazardous condition is detected, alarm activation and mechanical responses shall occur automatically without user intervention. Emergency procedures shall remain active until the hazardous condition is resolved.

### **S5 Maintainability – Modular embedded design**

The system architecture shall support modular separation between sensor monitoring, actuator control, display handling, and alarm management. Modifications to one subsystem shall not require significant restructuring of unrelated components. The design shall promote clarity and separation of functional responsibilities.

### **S6 Power stability and hardware robustness**

The system shall operate within the electrical limits of the hardware components. High-power devices such as servo motors shall not compromise system stability. The system shall manage actuator activation to prevent hardware overload or unexpected power failure.

### **S7 System documentation quality**

The system design shall be documented clearly and consistently. Documentation shall include structured requirement specifications, architectural descriptions, and system behavior explanations to facilitate understanding, evaluation, and future development.

## References

1. Nielsen, J. (1993). *Usability Engineering*. Morgan Kaufmann.
2. Sommerville, I. (2016). *Software Engineering* (10th ed.). Pearson.